

Introduction to Probability Distribution Fitting

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Lecture Overview

Distribution fitting

- Finding the best-fitting theoretical probability distribution for the observed data

Three approaches covered in this lecture:

- **MASS**: `fitdistr()`
- **fitdistrplus**: `fitdistr()`
- **gamlss**: `fitDist()`

Notes:

- No need to specify distribution function for the last approach, i.e., `fitDist()`
- Introduction and overview to a very broad field of research

Introduction

Empirical work often requires understanding of the underlying distribution of data:

- Distribution of corn yields in a particular county based on observations to calculate the probability of getting a yield below a certain threshold, e.g., for crop insurance purposes
- Wind speed distribution at a particular location for construction of a wind farm: Electricity production is not possible below and above a certain wind speed

Estimation of one or more parameters characterizing a probability distribution function

Weibull: Random Data Generation

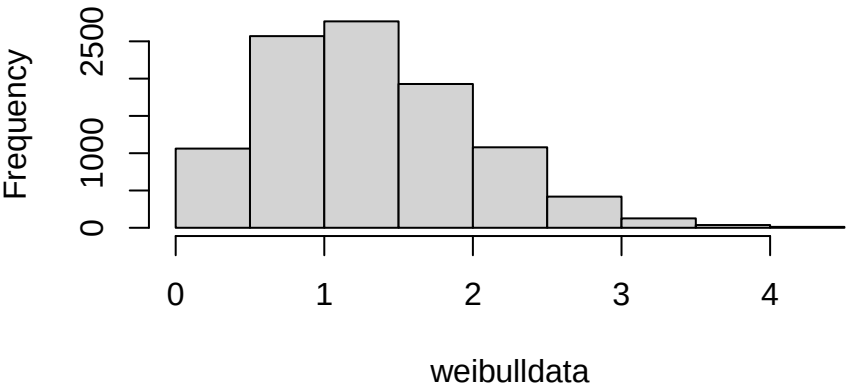
Random generation of data ($N=10000$) following a Weibull distribution with two parameters:

- Shape: $k = 2$
- Scale: $\lambda = 1.5$

```
weibulldata = rweibull(10000,2,1.5)
```

Weibull: Histogram

Histogram of Weibull Data



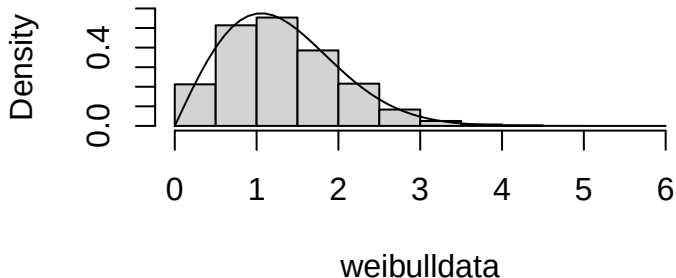
Weibull: Distribution Fitting with `fitdistr`

```
weibullpara    = fitdistr(weibulldata,densfun="weibull",  
                           lower=c(0,0))  
shape          = weibullpara$estimate[1]  
scale          = weibullpara$estimate[2]  
c(shape,scale)  
  
##      shape      scale  
## 2.002244 1.492127
```

Weibull: Observed Data and Estimated Distribution

```
hist(weibulldata,freq=FALSE,ylim=c(0,0.6),xlim=c(0,6))  
range          = seq(0,6,0.1)  
lines(range,dweibull(range,shape,scale))
```

Histogram of weibulldata



Distribution Fitting Steps

General steps (see [Fitting Distributions with R](#) by Vito Ricci for more information)

- ① General hypothesis about candidate distributions, e.g., discrete vs. continuous, entire real number line vs. positive numbers only
 - Histogram as a valuable first approach
- ② Parameter estimation
 - Example: Calculating shape and scale parameters of the Weibull distribution or mean and variance for a Normal distribution
- ③ Goodness of fit

Starting point for an overview of various probability distributions: [List of probability distributions](#)

Meridian Hills: Possible Distributions

Meridian Hills home values:

- Source: <https://jrfdumortier.github.io/dataanalysis/>
- 101 home values in the Meridian Hills neighborhood in Indianapolis
- Scaling of data to measure home values in \$1000

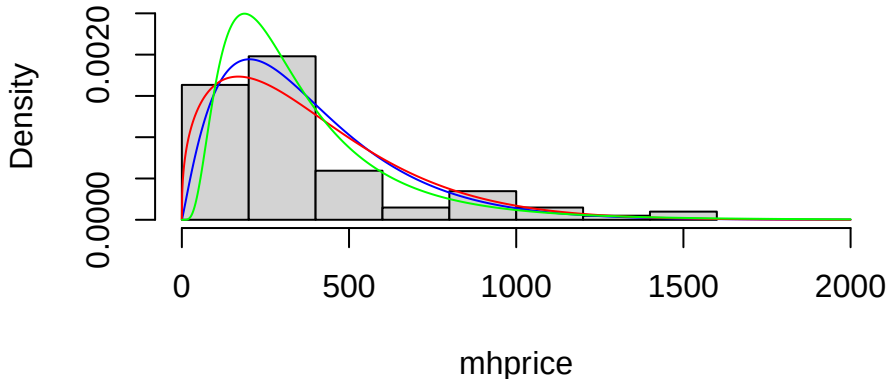
Candidate distributions:

- Gamma distribution: Shape and scale parameter
- Weibull distribution: Shape and scale parameter
- Log-normal distribution, i.e, $Y = \ln(X)$ has a normal distribution: μ and σ

```
mhprice      = mh1$price/1000
mhgamma      = fitdistr(mhprice,"gamma")
mhweibull    = fitdistr(mhprice,"weibull",lower=c(0,0))
mhlognormal  = fitdistr(mhprice,"log-normal")
```


Meridian Hills: Histogram II

Meridian Hills



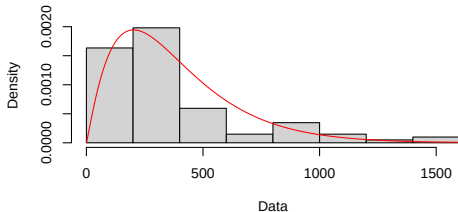
Meridian Hills: Setup for `fitdist()`

Use of the function `fitdist()` from the package `fitdistrplus`

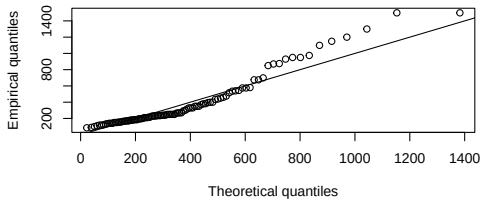
```
mhprice      = mh1$price/1000
mhgamma      = fitdist(mhprice,"gamma",lower=c(0,0))
mhweibull    = fitdist(mhprice,"weibull",lower=c(0,0))
mhlognormal  = fitdist(mhprice,"lnorm",lower=c(0,0))
```

Meridian Hills: Gamma Distribution

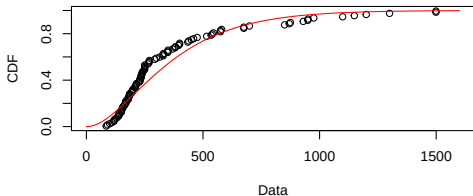
Empirical and theoretical dens.



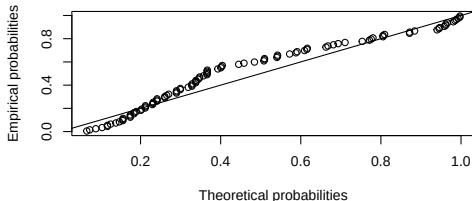
Q-Q plot



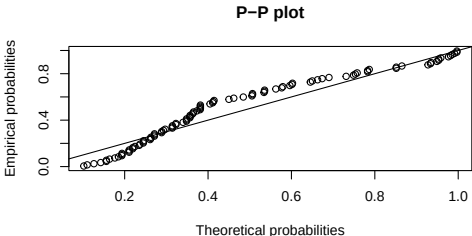
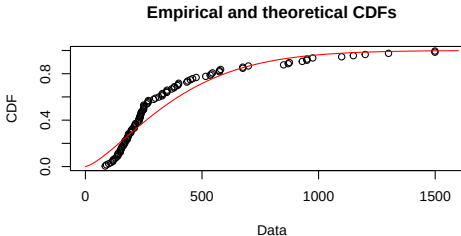
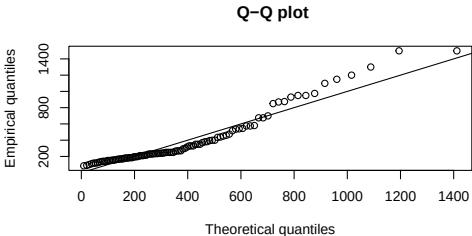
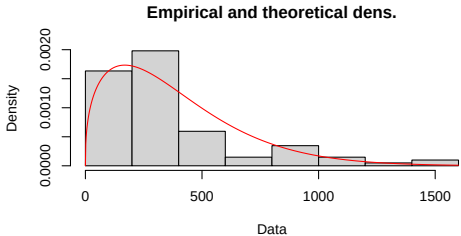
Empirical and theoretical CDFs



P-P plot

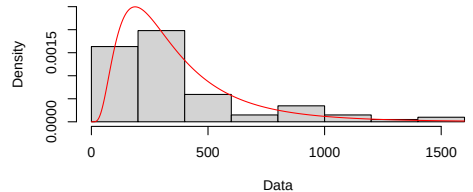


Meridian Hills: Weibull Distribution

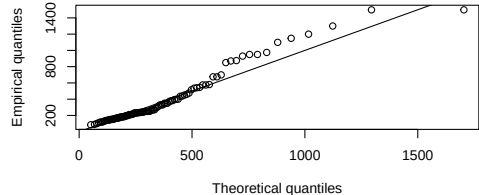


Meridian Hills: Log-Normal Distribution

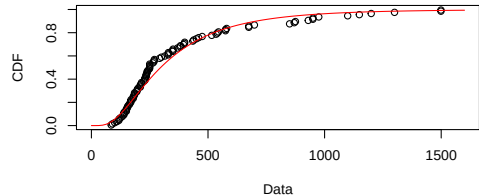
Empirical and theoretical dens.



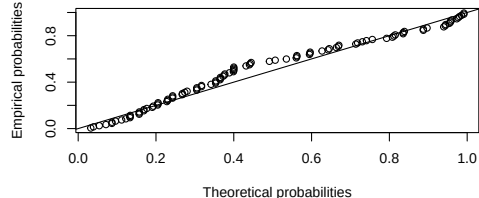
Q-Q plot



Empirical and theoretical CDFs



P-P plot



Ground Beef: Possible Distributions

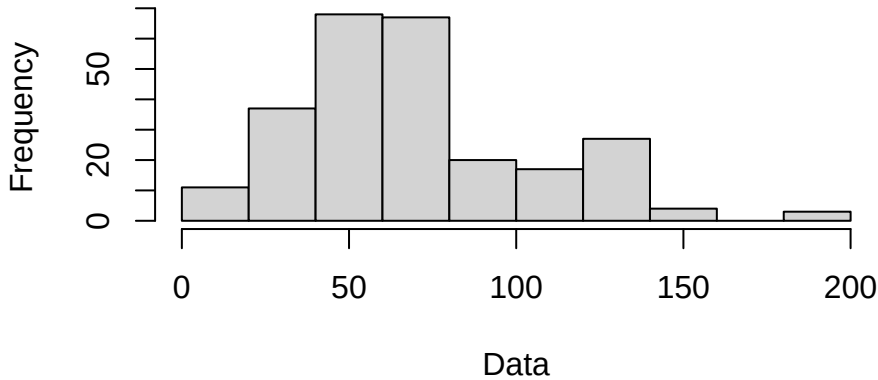
Second example using the function `fitdist()` package:

- Use of the data `groundbeef` associated with the package `fitdistrplus`:
Serving sizes collected in a French survey, for ground beef patties consumed by children under 5 years old.

```
data("groundbeef")
gbnormal      = fitdist(groundbeef$serving,"norm")
gbweibull     = fitdist(groundbeef$serving,"weibull")
gbgamma       = fitdist(groundbeef$serving,"gamma")
gblognormal   = fitdist(groundbeef$serving,"lnorm")
fitteddist    = list(gbnormal,gbweibull,gbgamma,gblognormal)
plotlegend     = c("Normal","Weibull","Gamma","Lognormal")
```

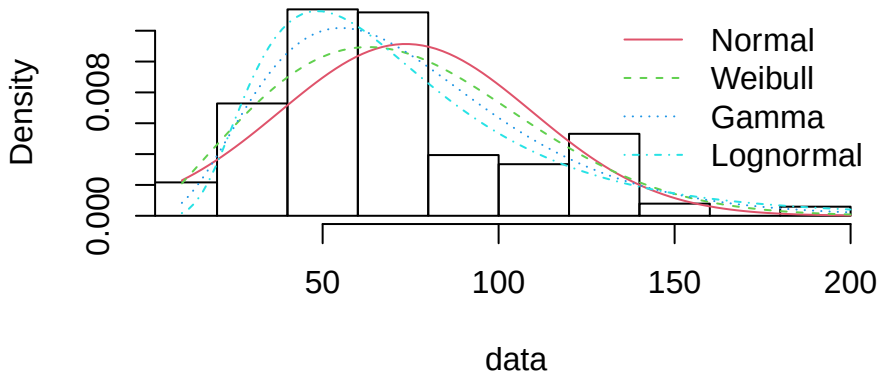
Ground Beef: Histogram

Ground Beef



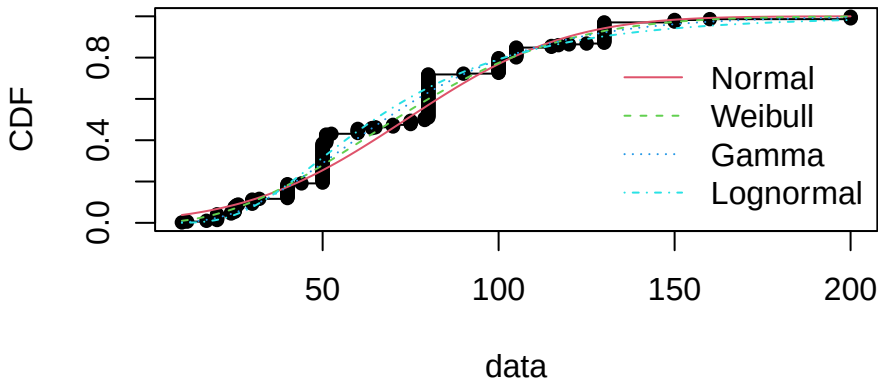
Ground Beef: Results I

Histogram and theoretical densities

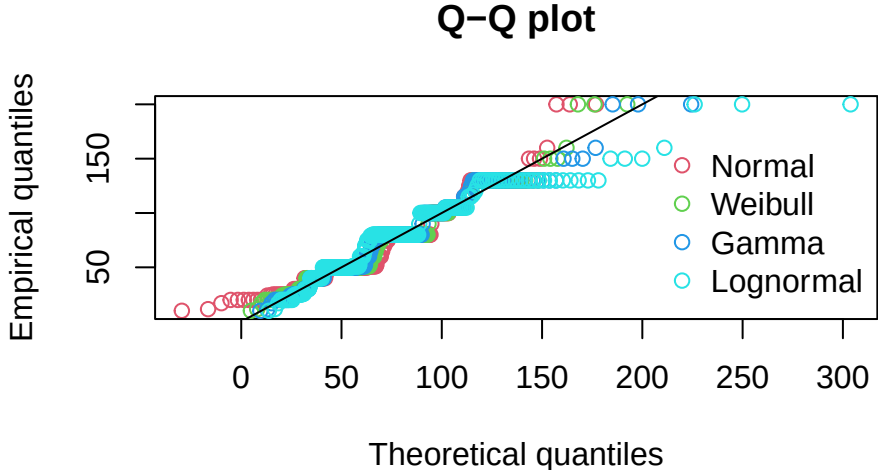


Ground Beef: Results II

Empirical and theoretical CDFs



Results: Q-Q Plot



Unspecified Distribution: `fitDist()`

Use of the function `fitDist()` from package `gamlss`

```
output = fitDist(mhprice,type="realplus")
```

```
output$family
```

```
## [1] "IGAMMA"          "Inverse Gamma"
```

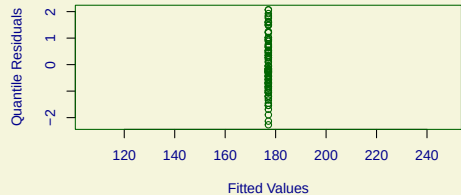
```
output$Allpar
```

```
##      eta.mu  eta.sigma
```

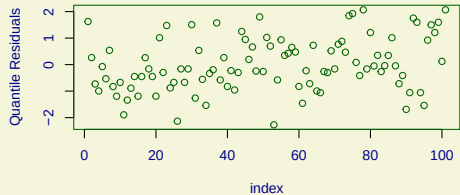
```
## 5.1768720 -0.4921408
```

Goodness of Fit with Inverse Gamma

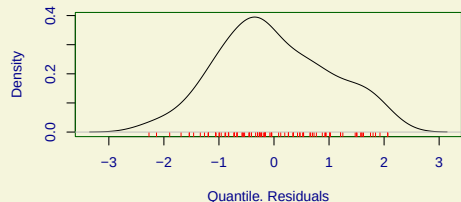
Against Fitted Values



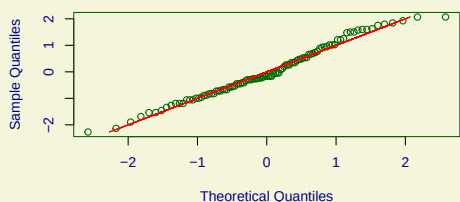
Against index



Density Estimate



Normal Q-Q Plot

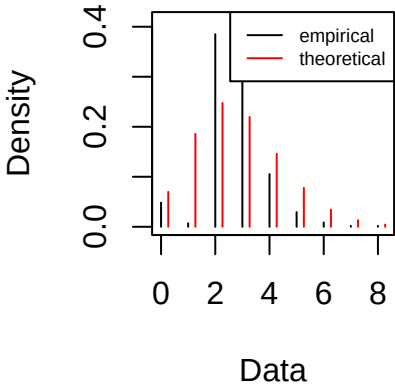


EV Data

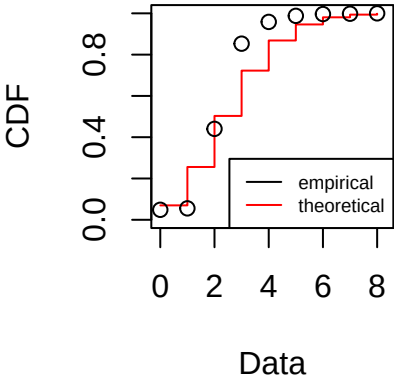
```
evpoisson = fitdist(evdata$numcars,discrete=TRUE,distr="pois")  
evnbinom  = fitdist(evdata$numcars,discrete=TRUE,distr="nbinom")
```

EV Data: Results Poisson

Emp. and theo. distr.

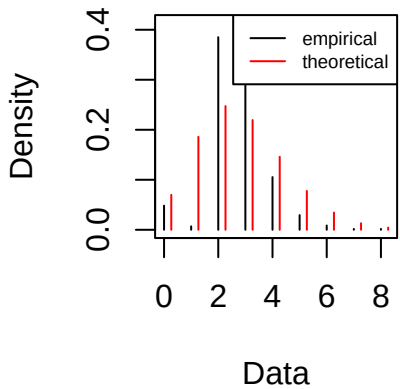


Emp. and theo. CDFs



EV Data: Results Negative Binomial

Emp. and theo. distr.



Emp. and theo. CDFs

